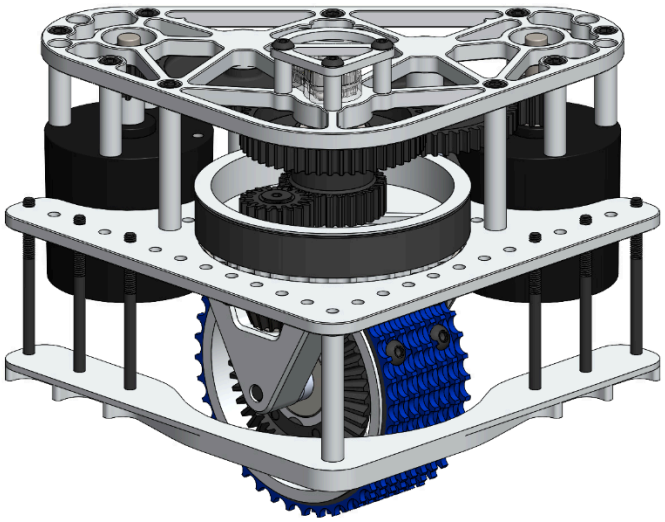
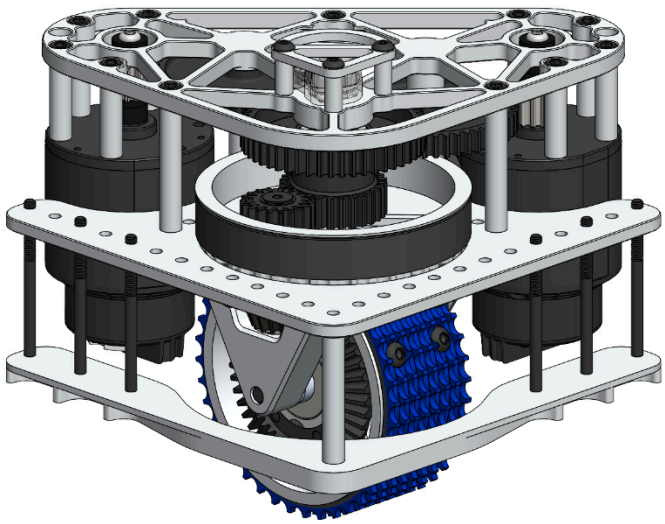
SWERVE DRIVE SPECIALTIES

MK4i Swerve Module
Assembly Guide

v20241114a





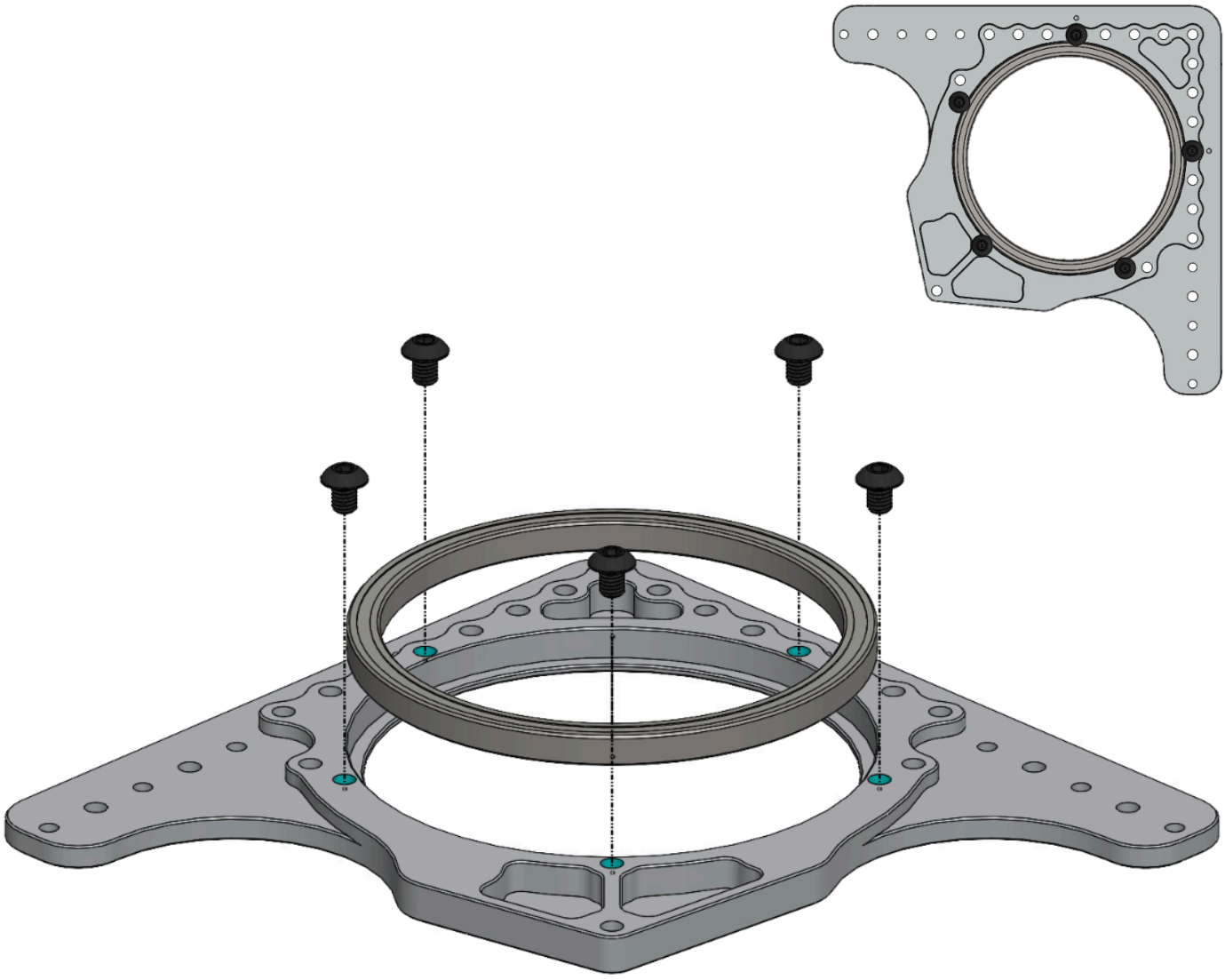
Recommendations:

Use Loctite 243 Threadlocker on all bolts, excluding the screws used to mount the encoder

Lubricate gears with white lithium or Red “N” Tacky grease after assembly

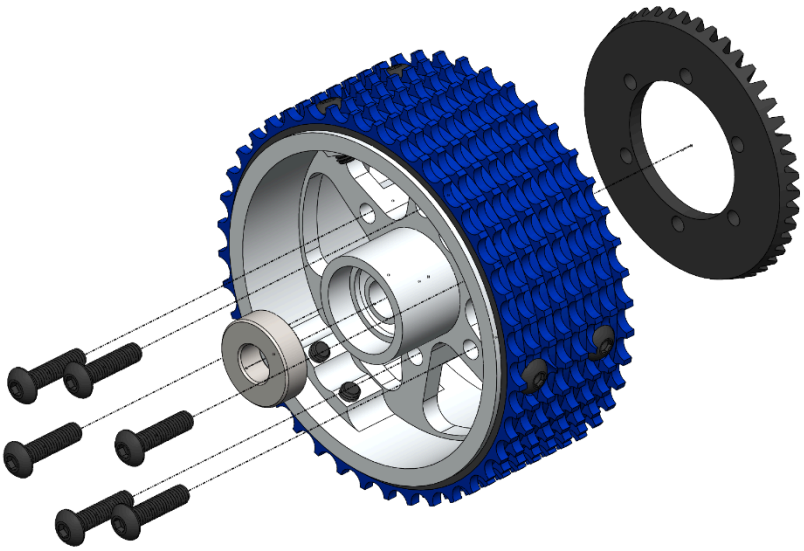
Step 1: Main Bearing Mounting

Secure with five #10-32 X .25 Button Head Screws. Nonobvious screw positions are indicated by adjacent drilled dots.

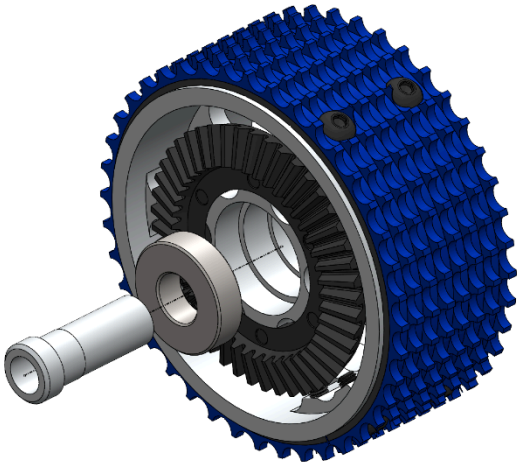


Step 2: Wheel Assembly

The MK4i Module uses the 4"D X 1.5"W Billet Wheel from SDS. Tread comes pre-installed. Mount 45t Bevel Gear with six #10-32 X .75 Button Head Screws. Insert .5” ID Bearing and Wheel Spacer on 45t Bevel Side, insert .375” ID Bearing on opposite side.







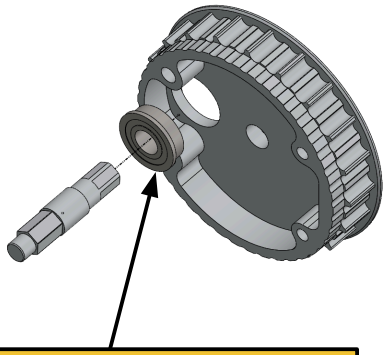


Step 3: Intermediate Shaft Assembly

Install Bearing

For V1/V2 Intermediate Shaft

Insert FR6ZZ-10.25mmID *Flanged* Bearing and Shaft



IMPORTANT
If using the flanged bearing (for V1/V2 shaft), ensure its flange is on the **bottom** of the Base Pulley

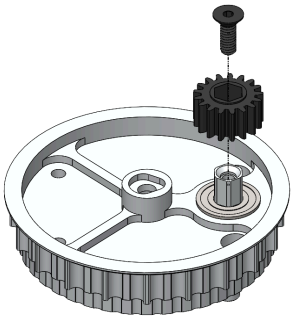
For V3 Intermediate Shaft

Insert 6802ZZ Bearing (*not* flanged) and Shaft

Secure Shaft and 20DP .375 Hex Bore Gear

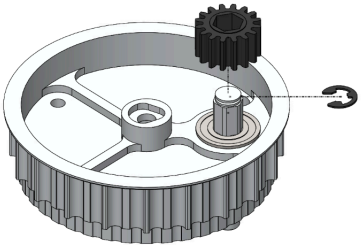
For V1 Intermediate Shaft

Install #12-24 X .5” Flat Head Screw



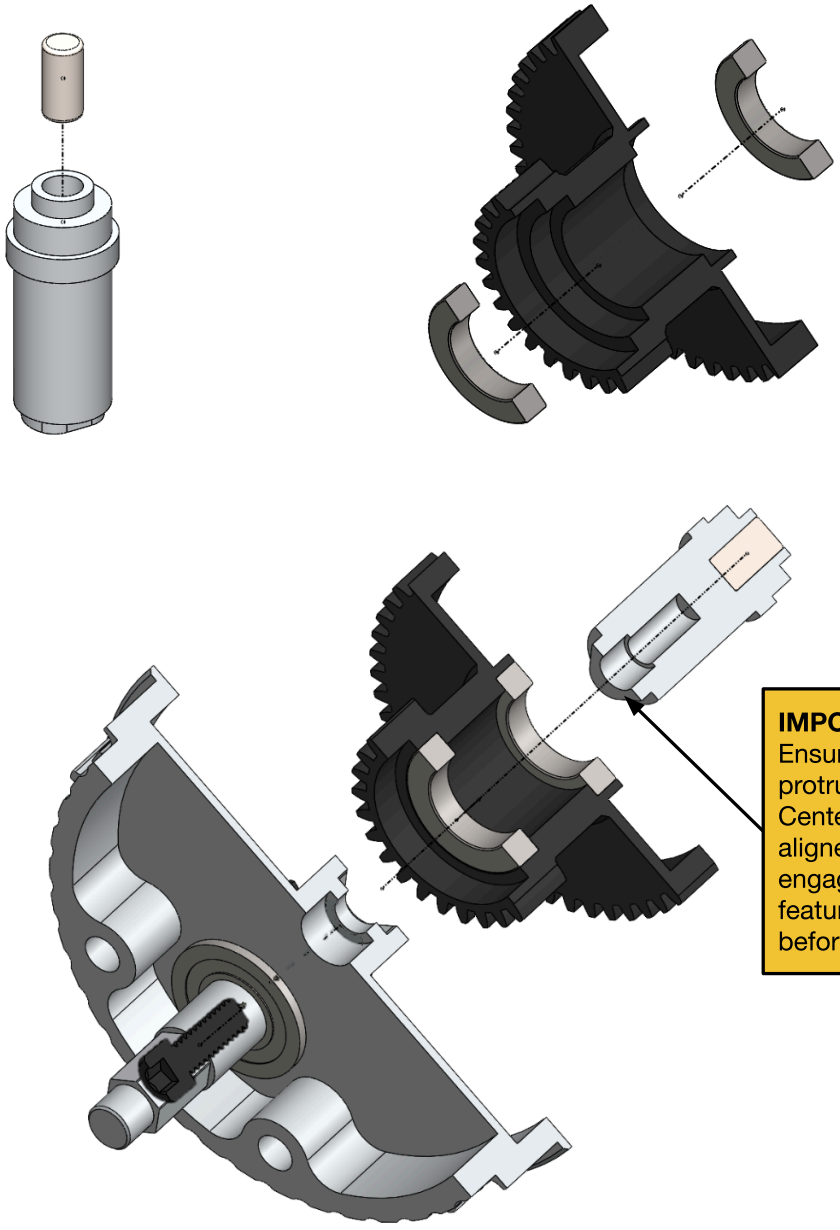
For V2/V3 Intermediate Shaft

Install E-Clip into Shaft’s groove



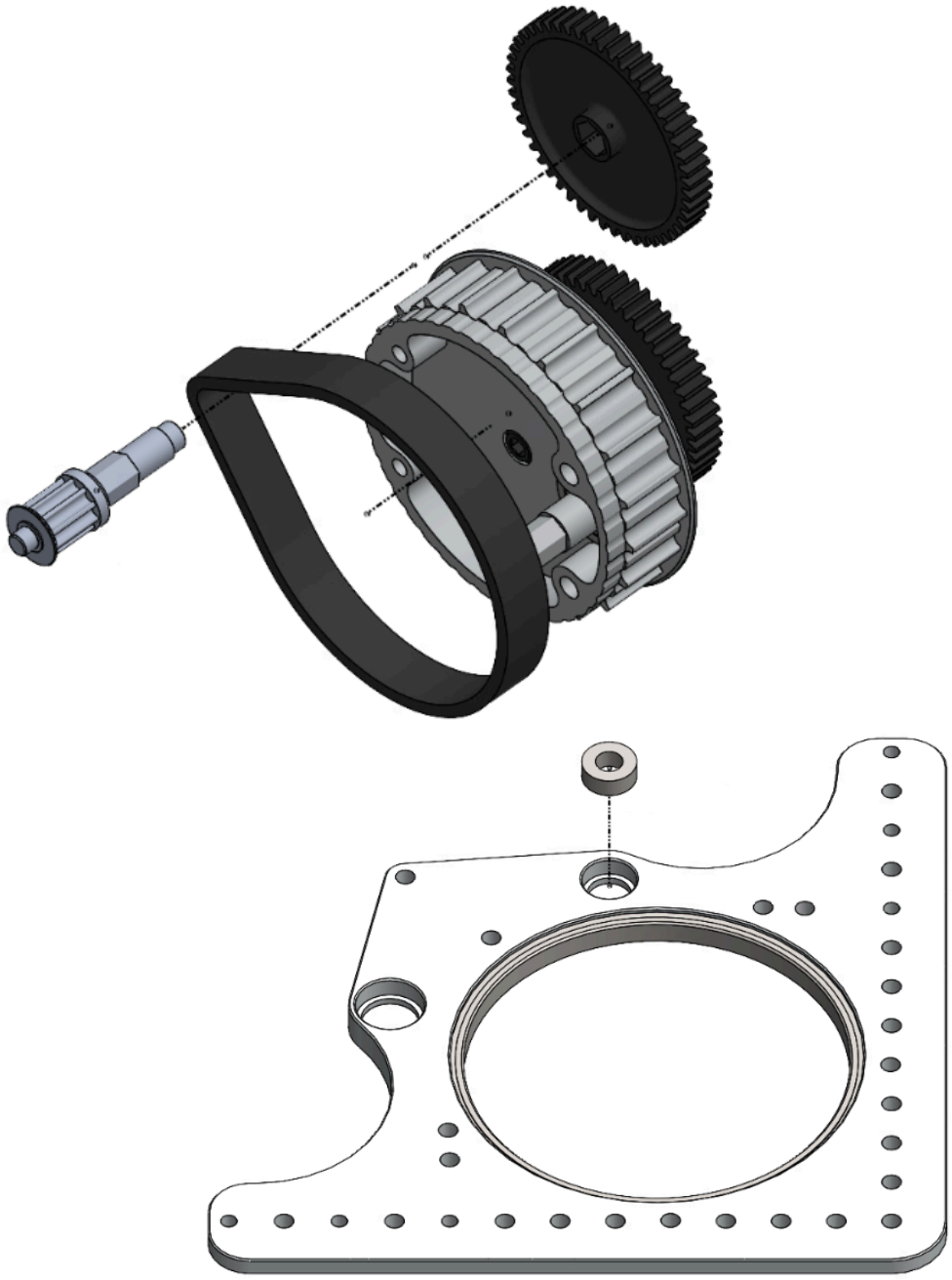
Step 4: Center Column Assembly

Insert encoder magnet (provided separately with encoder) into Center Column and retain with Loctite 609 Retaining Compound (wipe off excess), ensuring magnet is flush with top of Center Column. Insert two 6802ZZ Bearings into Double Gear and capture with Center Column and 1/4”-20 X .625” Socket Head Cap Screw.



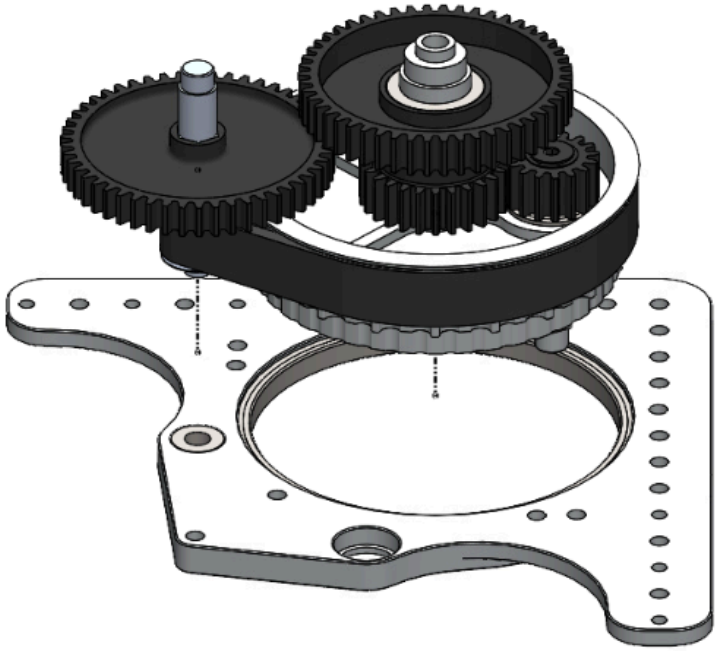
Step 5: Pulley Mounting

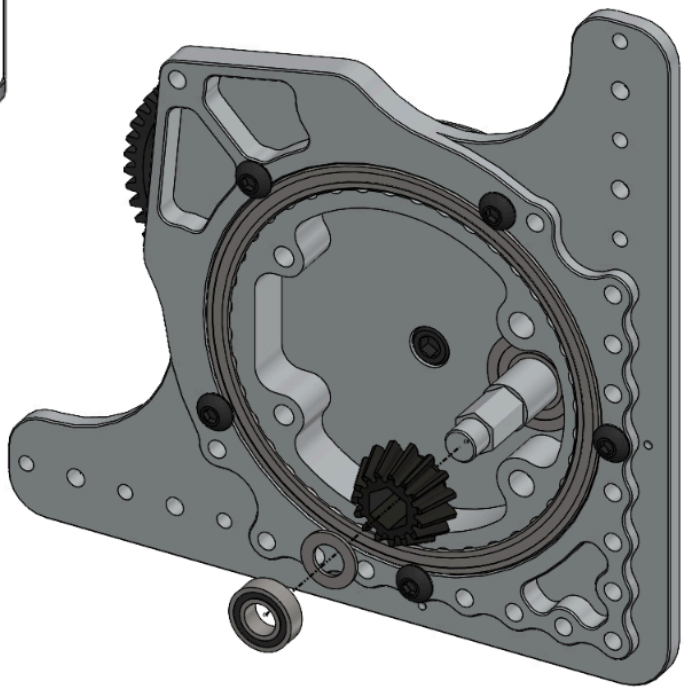
Capture Steering Belt between Steering Shaft and Base Pulley. Insert 20DP 50t gear with .375” Hex bore onto Steering Shaft and between Base Pulley and Double Gear. Insert R188ZZ Bearing in Main Plate before continuing with assembly.



Step 5: Pulley Mounting (Cont.)

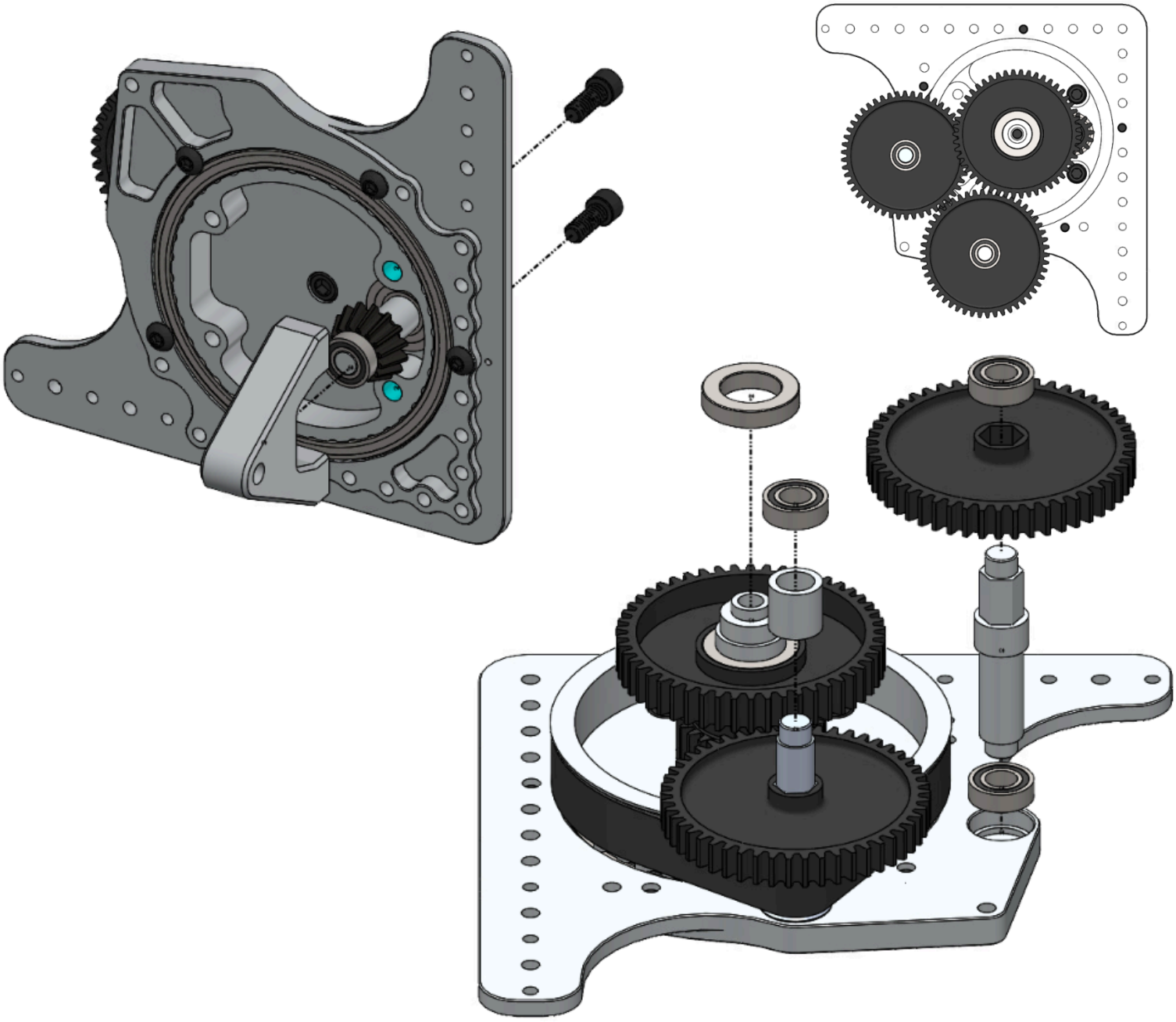
Insert Base Pulley Assembly, aligning Base Pulley with X-Contact Bearing and Steering Shaft with R188ZZ Bearing. Insert 15t Bevel Gear, followed by 1mm Shim and 688 Bearing.





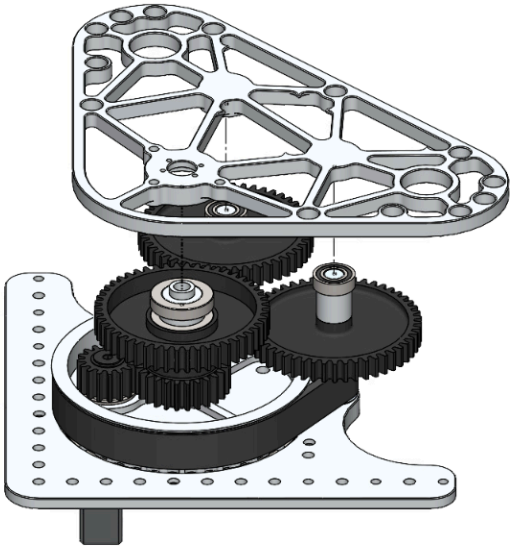
Step 6: Wheel and Motor Mount Prep.

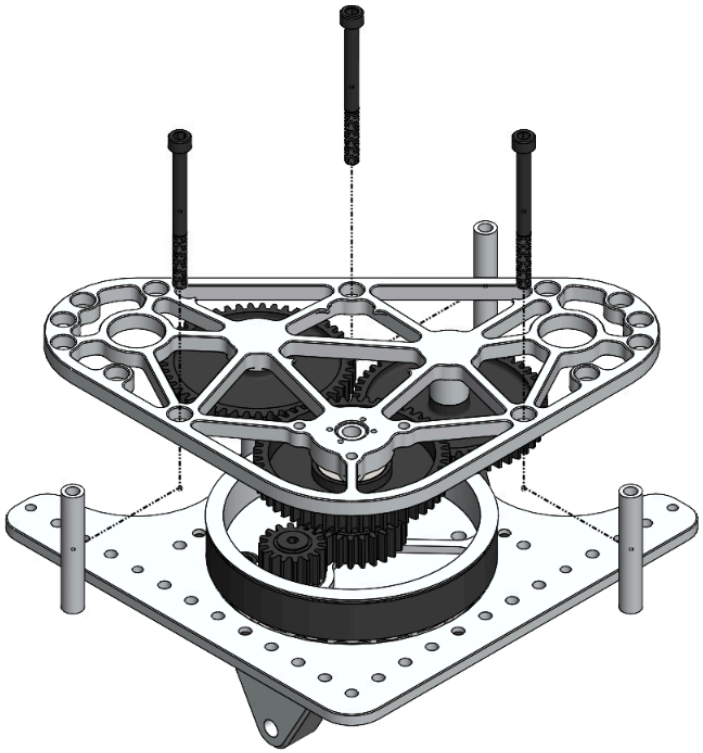
Attach Wheel Mount B with two 1/4-20 X .625 Socket Head Cap Screws. Place three 688 Bearings, 20DP 50t Gear, .5”OD .375”ID Spacer and 6802ZZ Bearing on respective shafts.

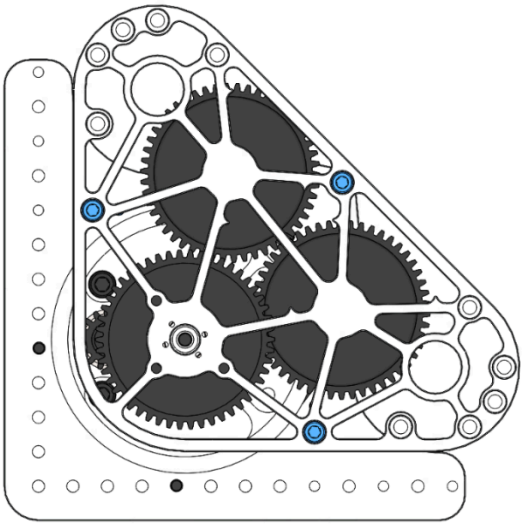


Step 7: Motor Plate Mounting

Line up the three bearing bores and attach Motor Plate. Be sure all three bearings are seated in the bearing bores and secure with three #10-32 X 2.25” screws through three #10 X 1.875” Spacers.







Step 8: Motor Configurations

Falcon 500 Motor:

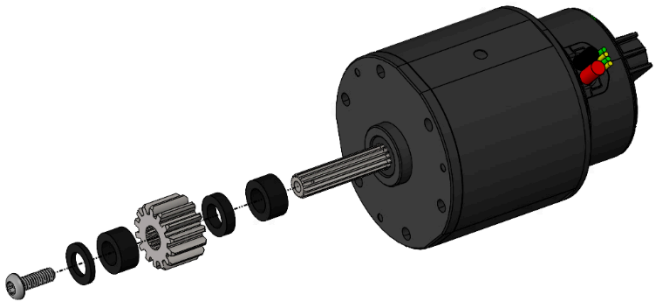
Use the acetal spacers that are provided with the Falcon 500 motors.

Drive Motor: .125” and .25” spacers before pinion, 20DP 14t Pinion, .25” and .0625” after. Secure with 8-32 X 1/2" truss head screw.

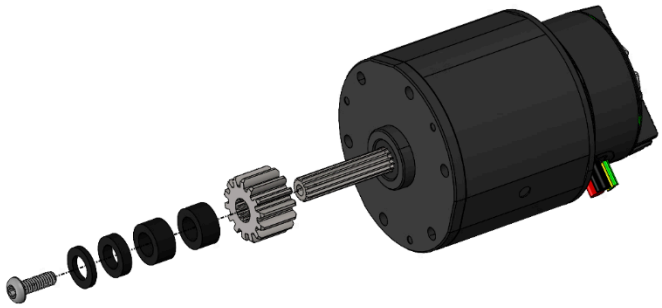
Steering Motor: 20DP 14t Pinion, 2 X .25”, .125”, .0625” spacers after. Secure with 8-32 X 1/2" truss head screw.

Assemble with medium strength retaining compound on retaining screw.

Drive



Steering



Step 8: Motor Configurations (Cont.)

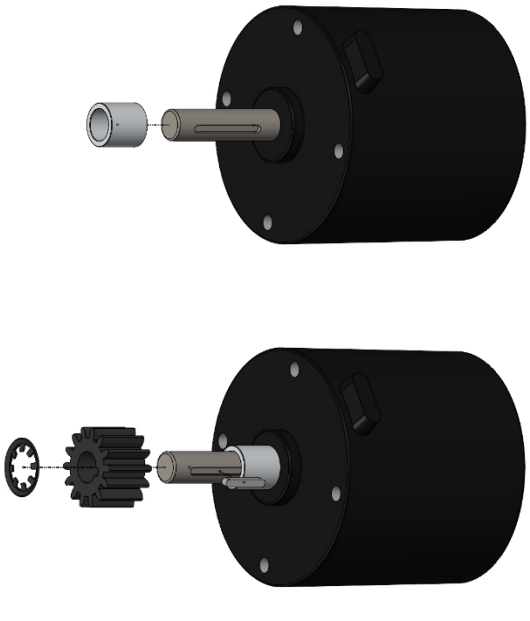
NEO and NEO Vortex Motors (8mm keyed shaft):

Drive Motor: .437” spacer before pinion, 20DP 14t Pinion, retaining ring after.

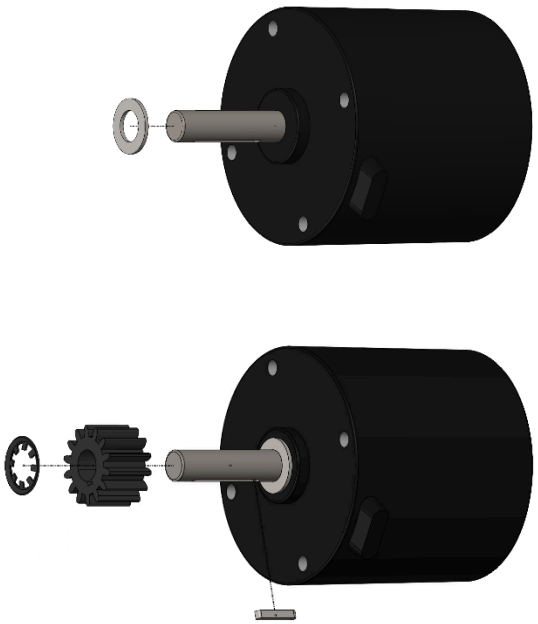
Steering Motor: 1mm spacer before pinion, 20DP 14t Pinion, retaining ring after.

Place spacers on NEO Motor shafts, followed by machine keys, gears, and retaining rings.

Drive



Steering



Step 8: Motor Configurations (Cont.)

Kraken X60 Motors:

Drive Motor: Install .125” spacer first, then 20DP 14t Pinion, and .375” spacer last. Secure with 10-32 X 1/2" Button Head Screw, using medium strength retaining compound.

Steering Motor: Install 20DP 14t Pinion first, then .125” spacer, and .375” spacer last. Secure with 10-32 X 1/2" Button Head Screw, using medium strength retaining compound.

IMPORTANT To prevent an interference between the motors and Main Plate, remove the cooling port screws from the motors, and then cover the ports with tape.

Drive Motor

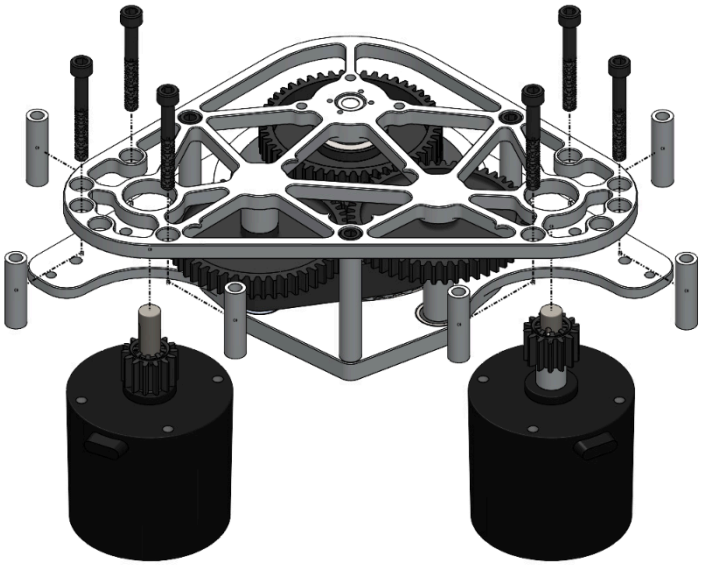


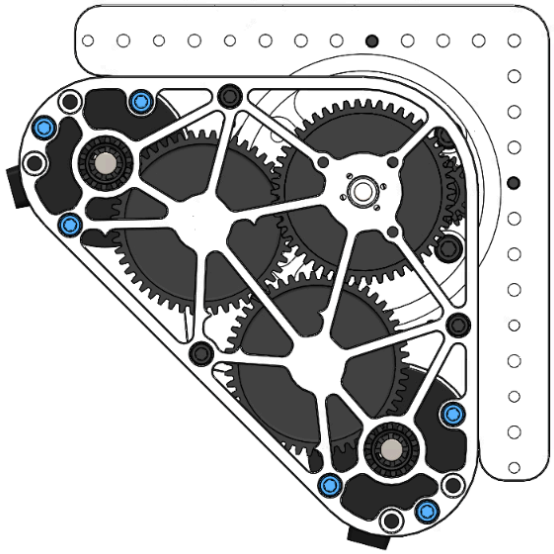
Steering Motor

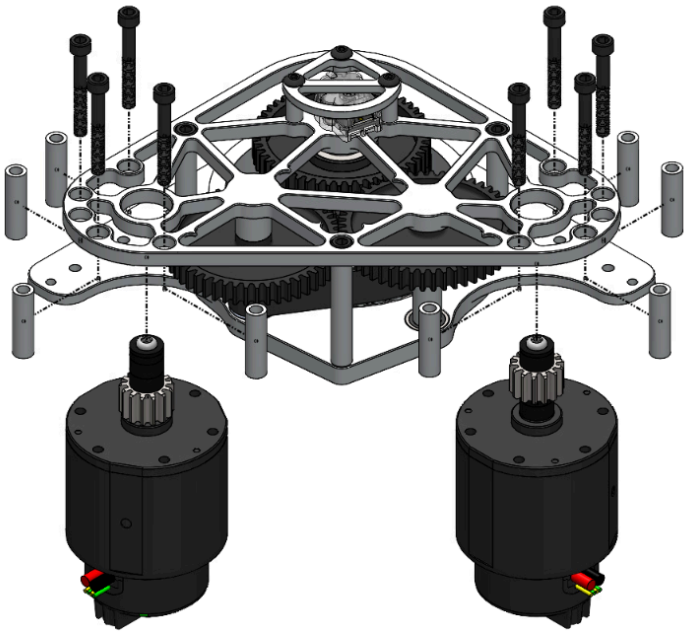


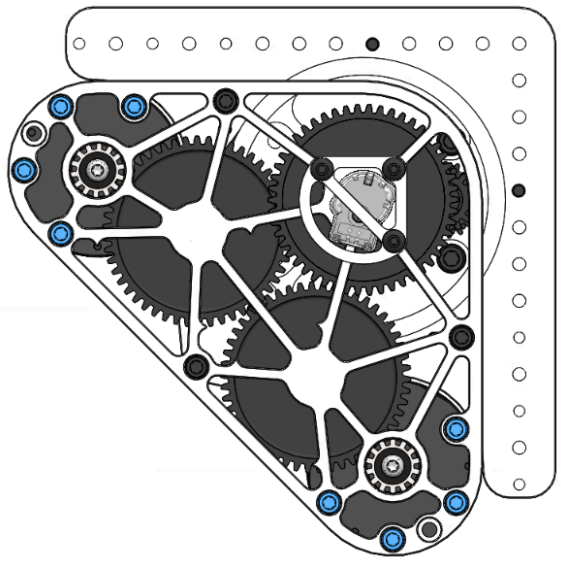
Step 9: Motor Mounting

Attach Falcon 500 Motors with four #10-32 X 1.5” screws and #10 X 1.125” spacers. For NEO, NEO Vortex, and Kraken X60 Motors, use three screws/spacers. **For NEO Vortex and Kraken X60 Motors,** install one Semicircle Spacer between each motor and three spacers (not shown).



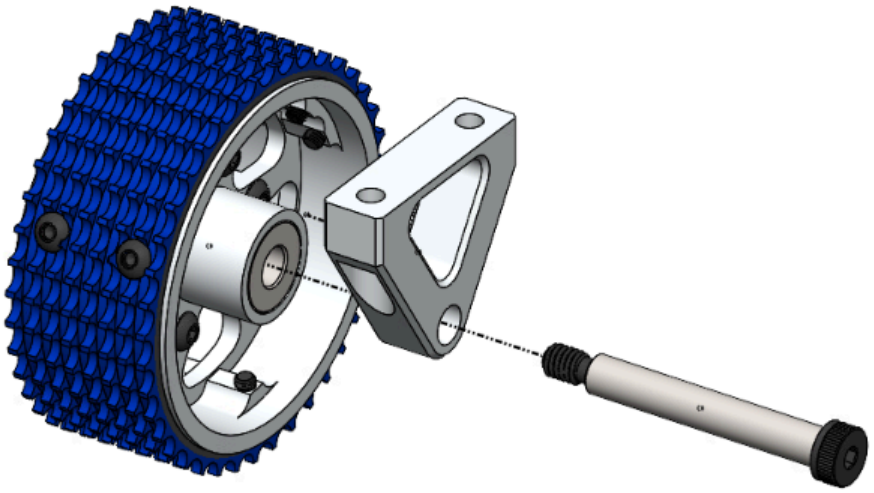


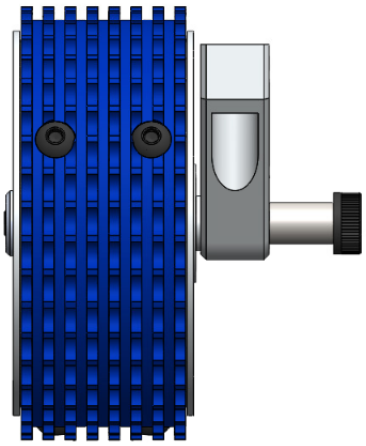


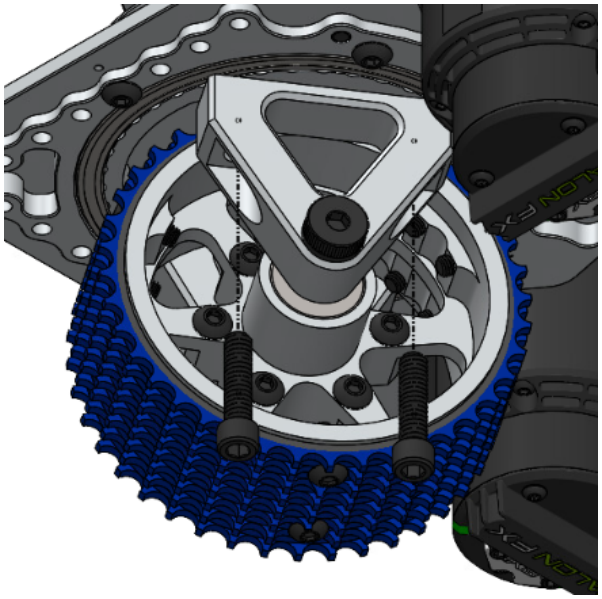


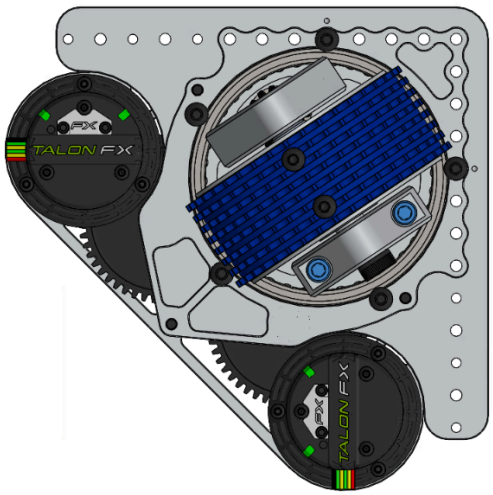
Step 10: Wheel Mounting

Insert Shoulder Bolt partially through Wheel Mount A with integral spacer facing wheel. Slide wheel subassembly module and screw Shoulder Bolt in partially. Secure Wheel Mount A with two 1/4-20 X 1” Socket Head Cap Screws. Screw in Shoulder Bolt.



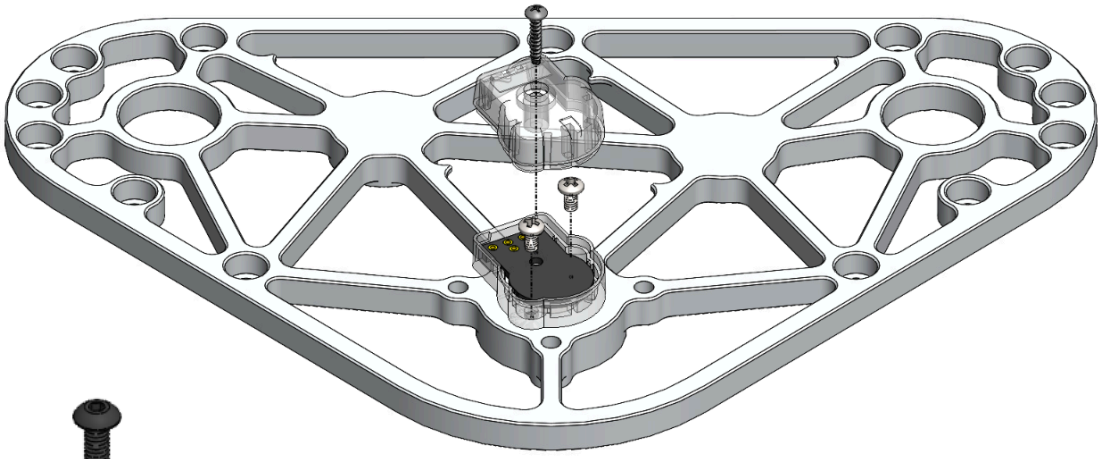


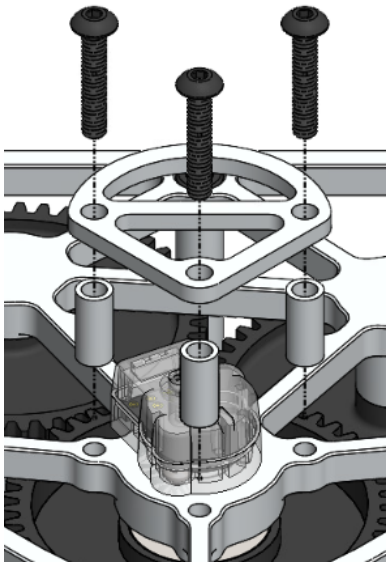


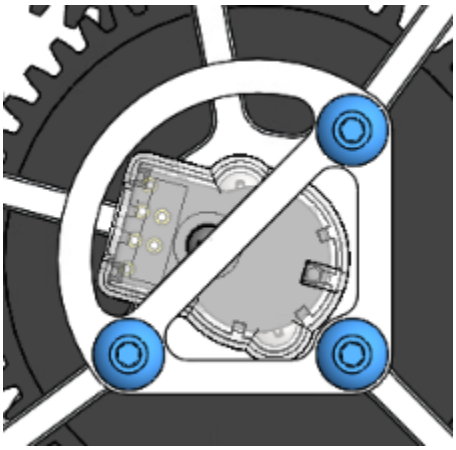


Step 11: Encoder Mounting

Begin by mounting housing base with both #3 – 48 screws included with the encoder (without thread locking compound) before inserting encoder circuit board (attaching data cable if using SRX Mag Encoder or soldering on CAN wires if using CANcoder). Choose the mounting holes that result with the wire direction you prefer. Place housing cap and follow CTR Electronics instructions: “DO NOT OVER TIGHTEN THE 2-28 SCREW AS THIS MAY RESULT IN PERMANENT DAMAGE TO THE HOUSING. HAND TIGHTEN UNTIL RESISTANCE IS FELT.” Cable tie encoder wires to Motor Plate using machined keeper features. Finally mount machined Encoder Enclosure using three provided 8-32 X .875” Button Screws and three #8 X .531” Spacers.







Step 12: Module Mounting

Attach final #10 X 1.875” spacer to module with #10-32 X 2.25” Socket Head Cap Screw. Mount to box tube with six provided 2.5” Socket Head Cap Screws (#8-32 or #10-32, depending on swerve module configuration).

IMPORTANT Lubricate gears with white lithium or Red “N” Tacky grease before usage, and regularly with use.

